

# STEREO AMPLIFIER Using TDA7052 ICs

## For PORTABLE DEVICES

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Portable devices like personal computers, MP3 players, sound recorders, telephones and audible sensors are frequently in need of additional amplifiers to produce sufficient sound. These audio amplifiers should work with

batteries (4.5V, 6V, 12V, etc) and USB interfaces (+ 5V). These should have low quiescent current and produce at least 2x0.5W of output power. This output power is enough for listening in a small room with several people.

### Circuit and working

Fig. 1 shows the circuit diagram of a simple stereo amplifier with two TDA7052 ICs (IC1 and IC2) that can be used for portable devices.

TDA7052 is a 1.2W mono audio power amplifier. It has a wide power supply range (3V to 18V) and can work from USB interfaces. Typical quiescent current of TDA7052 is 4mA, which makes it appropriate for battery operations. Gain of each channel of the circuit is 39dB (around 90). Typical input resistance of TDA7052 is 100-kilo-ohm, and the IC is suitable for a lot of high-impedance signal sources.

The circuit has six switches (S1 through S6). When S1 and S2 are open, input impedance of the circuit is more than 500-kilo-ohm and the amplifier can be used with electrical guitars, high-impedance sensors and signal sources.

When S3 and S4 are closed, input resistance is reduced and, at the same

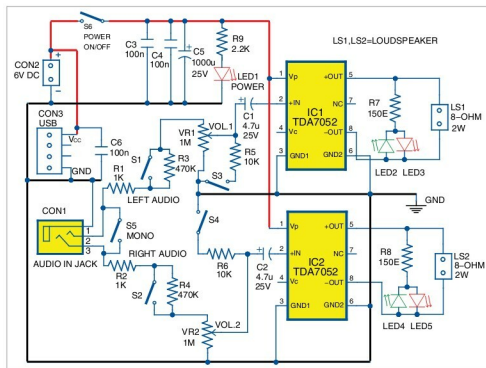


Fig. 1: Circuit diagram of the stereo amplifier for portable devices

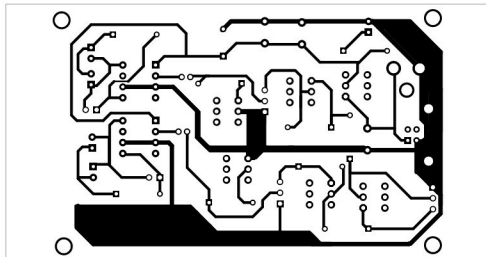


Fig. 2: Actual-size PCB layout of the stereo amplifier for portable devices

### PARTS LIST

|                                              |                            |
|----------------------------------------------|----------------------------|
| <b>Semiconductors:</b>                       |                            |
| IC1, IC2                                     | - TDA7052 audio amplifier  |
| LED1-LED5                                    | - 5mm LED                  |
| <b>Resistors (all 1/4-watt, ±5% carbon):</b> |                            |
| R1, R2                                       | - 1-kilo-ohm               |
| R3, R4                                       | - 470-kilo-ohm             |
| R5, R6                                       | - 10-kilo-ohm              |
| R7, R8                                       | - 150-ohm                  |
| R9                                           | - 2.2-kilo-ohm             |
| VR1, VR2                                     | - 1-mega-ohm potentiometer |
| <b>Capacitors:</b>                           |                            |
| C1, C2                                       | - 4.7µF, 25V electrolytic  |
| C3, C4, C6                                   | - 100nF ceramic disk       |
| C5                                           | - 1000µF, 25V electrolytic |
| <b>Miscellaneous:</b>                        |                            |
| S1-S6                                        | - On/off switch            |
| CON1                                         | - Stereo jack connector    |
| CON2                                         | - DC jack                  |
| CON3                                         | - Type-B USB connector     |
| LS1, LS2                                     | - 8-ohm, 2W loudspeaker    |